Shetty

by Dhanush G

Submission date: 10-Feb-2025 04:18PM (UTC+0530)
Submission ID: 2584564328
File name: ISE_Assignment_extended_version.docx (57.03K)
Word count: 3046
Character count: 19876

PROJECT REPORT

ON

Software Development Lifecycle (SDLC) Analysis of HMS

A comparative study of different models in relation Health Management System (HMS)

Submitted to

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(Off-Campus Centre, Nitte Deemed to be University, Nitte - 574 110, Karnataka, India)

In partial fulfilment of the requirements for the award of the

Degree of Bachelor of Technology

In

INFORMATION SCIENCE AND ENGINEERING

by

Dhanush G Shetty NNM23IS049

Under the guidance of

Dr. Jason Elroy Martis, Associate Professor,

Department of Information Science and Technology,

NMAM Institute of Technology. Nitte Karnataka, India

NITTE (Deemed to be University) | NMAM INSTITUTE OF TECHNOLOGY

Assignment-1: Software Process Models and Requirements Engineering

Unit-I: Introduction to Software Development and Requirements Engineering

Problem Statement
Analyze the software development lifecycle (SDLC) of real-world system by conducting a comparative study of process models and their impact on requirements management.

Case-Study: Healthcare Management System (HMS)
The HMS represents a large-scale Healthcare Management System serving healthcare facilities by integrating administrative operations with clinical requirements. The system handles necessary duties that include maintaining patient records as well as appointment scheduling combined with billing administration and inventory tracking and report production. Patient data stays safe using HIPAA and other protective security standards that the system follows.

Comparison and Analysis of Software Process Models

1. Incremental Development Model

The Incremental Development Model applies sequential methods to develop operational system components which appear at every advancement level. This development model works to provide essential features with operational functionality prior to final design evaluation and enables continuous improvement together with new idea generation.

Suitability for HMS:

- **Functional Requirements**: Healthcare organizations utilize the model's functional requirements to deploy necessary features one by one starting from patient registration and moving through appointment scheduling and billing procedures. The critical set of functionalities should be delayed first and then introduced early as ready-to-use capabilities for health care providers.
- **Non-Functional Requirements**: Various functionality elements integrate at each development stage through the progressive approach under performance and security requirements and more. Basic incremental deployments will initiate the security system before its complete HIPAA implementation phase during a later development stage. Time strengthens the system as developers fulfil non-functional requirements at every development stage through this method.
- **Risk and Change Management:** The incremental Model allows Risk and Change Management to operate efficiently since programmatic changes become part-built into the system development process. Each stage of the system development enables user feedback that gets incorporated into future development sections to minimize important defects. During development processes the implementation structure facilitated a smooth integration between regulations and legal requirements.

- **Time and Cost Constraints:** Little Time and Money Constraints Lead to Early Delivery of Critical System Elements through Project Breakdown. Healthcare providers start using essential functions immediately through this method so they can handle budgeting and scheduling needs.

Advantages:

- Adaptable to the changes in the requirement.

Disadvantages:

- As more increments are added, it might become difficult to integrate them into system.
- Requires a great deal of planning and management.

Detailed Application to HMS:

The Increment development Model delivers excellent results in Healthcare Management System development because it enables the delivery of essential functions such as patient registration and appointment management from the start of the plan. The system can be implemented incrementally which allows the healthcare providers to start using basic features quickly while additional functions become available during the development process. With incremental development stakeholders obtain the chance to provide user feedback since they can contribute their responses at different stages of system development to ensure it maintains alignment with their expectations.

---

2. Spiral Model

The Spiral Model provides iterative risk analysis alongside prototyping features that make it an appropriate solution for difficult and uncertain projects.

Suitability for HMS:

- **Functional Requirements:** The method enables comprehensive testing of essential patient confidentiality measures along with regulatory standards.
- **Non-Functional Requirements:** The system continues to advance performance and security features along with reliability through a cyclical improvement process.
- **Risk and Change Management:** The system continues to advance performance and security features along with reliability through a cyclical improvement process.
- **Time and Cost Constraints:** The initial costs rise because of risk analysis and prototyping but these steps decrease both long-term risks and expensive rework needs.

Advantages:

- Utmost importance on risk management.
- Contains frequent changes during development.

- Delivers a high-quality product through continuous validation.

Disadvantages:

- Requires expert human resources for risk analysis and prototyping.
- Laborious due to its iterative nature.

Detailed Application to HMS:
An HMS benefits from the Spiral Model because this methodology involves repetitive cycles alongside risk-oriented solution development. The prototyping of regulatory requirements such as HIPAA takes place during early development iterations to eliminate later expenses for rework. The model begins by implementing risk evaluation which enables developers to handle essential factors including data protection and system dependability immediately. Through continuous validation of requirements combined with solutions the Spiral Model enables a system that satisfies multiple healthcare provider and patient needs. System scalability and user interface optimization issues can be predicted through iterative development hence developers can create solutions in advance.

---

3. Waterfall Model

The Waterfall Model functions through a linear sequence of project stages therefore it works best for well-defined projects without significant changes.

Suitability for HMS:

- **Functional Requirements:** The system produces effective results for well-defined requirements that show minimal likelihood of change such as basic patient registration.
- **Non-Functional Requirements:** A guarantee exists for thorough documentation and careful planning to make systems expandable for future development and fulfil all essential regulatory needs and compliance standards.
- **Risk and Change Management:** The system becomes rigid to modifications after development starts.
- **Time and Cost Constraints:** Project planning has definite schedules and spending limits yet a mid-development change will result in high expenses.

Advantages:

- Straightforward and direct.
- Projects that maintain well-documented system will achieve long-lasting maintainability through easily accessible reference materials for upcoming system updates.
- Works well for steady requirements.

Disadvantages:

- Addition of required changes after development commencement proves difficult for this system.
- Delayed delivery of the functional product.
- Later detection of problems becomes more likely when operating during the development phase.

Detailed Application to HMS:
The waterfall Model provides successful usage for creating uncomplicated small HMS components while requirement alteration remain low. The proper documentation stage should precede construction in basic reporting module development when using the waterfall model. The arrowhead model does not offer effective solutions for developing systems with user-oriented capabilities or dynamic functionalities. New requirements during development stages pose significant challenges to the linear design structure of the model.

Requirements Engineering Process

Simplified Requirements Document

Functional Requirements:

1. **Patient Registration and Management:** The system allows smooth patient record management with its features for adding patients while also handling updates and deletions in a secure manner.
2. **Appointment Scheduling:** The tool enables patients to schedule appoints in various convenient ways as well as change or cancel previous appointments which enhances both operational simplicity and user satisfaction.
3. **Billing and Payments:** The system conducts billing activities and payment operations and manages insurances claims efficiently for accurate financial processing.
4. **Inventory Management:** The system tracks hospital medical supplies and equipment while recording this data to avoid stock deficiencies and sustain operational procedures.
5. **Reporting:** the system produces detailed data reports that administrators alongside healthcare providers use to improve their decision-making abilities and measure organizational performance.

Non-Functional Requirements:

1. **Security:** Security requirements of HIPAA regulation receive top priority as the system implements security measures which protect sensitive patient information.
2. **Performance**: The system sustains performance levels with 10,000 active users at a time because it was designed for this workload capacity. The system provides optimized software and hardware components which enable simultaneous operations and maintains system performance quality at all times.

3. **Scalability:** further healthcare development needs are supported through the system design because it allows extra module integration including telemedicine services. The system provides flexible features that support growing market requirements by enabling operations to expand their service area while maintaining normal performance activities.
4. **Reliability:** the system operates 99.9% uptime reliability which ensures constant availability of critical healthcare functions.
5. **Usability:** A user-friendly interface has been designed for medical professionals and patients because it enhances system usability to create simple interactions along with brief practical learning experiences.

Requirements Elicitation Techniques

1. **Interviews:** Healthcare professionals including doctors, nurses and administrators require individual interviews for gathering specific day to day operational challenge and individual needs. The method enables the discovery of particular requirements which apply to specific contexts.
2. **Surveys and Questionnaires:** Well-structured surveys should be used to distribute to patients, caregivers and other end-users for gathering their input about desired system features and system usability while obtaining valuable feedback about their preferences. The method provides an effective way to receive feedback from numerous respondents.
3. **Workshops:** The health system will organize workshops that bring together IC staff, hospital leaders and healthcare providers to establish and rank system requirements through consensus-based planning.
4. **Observation:** System observation allows personnel t identify functional weaknesses in healthcare facilities so the Healthcare Management System (HMS) can effectively resole them. This real-world perspective helps refine requirements.
5. **Document Analysis:** The analysis of system documents and industry standards and HIPAA regulations will verify that the Healthcare Management System follow legal standards while integrating with other connected systems.

Requirements Validation Strategy

1. **Stakeholder Reviews:** Hospital staff members such as administrators and healthcare providers along with patient representatives will review collected requirements in continuous stakeholder review sessions. By engaging in discussions stakeholders can verify that their expectations have been met necessary demands are properly handled.
2. **Prototyping:** the development team will generate realistic prototypes of the system for stakeholders to touch and test. Though this method system developers can quickly locate requirements-based problems and confusion to refine specifications in a more accurate direction.
3. **Use Case Scenarios:** real-world scenarios need to be simulated to evaluate how well the system performs regarding its functional needs and non-functional specifications.

Such practical validation provides assurance regarding how well the system applies to real healthcare operational flows.

4. **Traceability Matrix:** Development of a traceability matrix should happen to show how requirements relate to design elements and test cases. The establishment of this structure conducts a comprehensive review of every requirement while development and testing progresses to avoid critical feature omissions.

Potential Challenges

1. **Diverse Stakeholder Needs:** The challenge involves merging clashes user group requirements ranging from doctors to nurses to administrators to patients. The different interests between medical staff and patients lead doctors to need quick patient record retrieval yet patients require straightforward scheduling methods. Completing these needs involves successful communication processes and group-based decision-production methods.
2. **Changing Regulations:** Medical practices operate under maximum regulation through evolving requirements including HIPAA holds authority. Implementing new regulatory standards during application development causes development delays while necessitating extra resources needed to achieve compliance at acceptable project timelines.
3. **Ambiguity in Requirements:** System outcome failure emerges from poorly described or unfocused requirements which cause developers to misunderstand what users want or systems need to achieve. The resolution of this challenge needs detailed documentation together with stakeholder approval followed by repeated refinement until clarity emerges.
4. **Resource Constraints:** The development process suffers from resource limitations which include restricted access to qualified personnel together with constrained project time periods alongside restricted budgets. The use of effective resource distribution schemes together with essential feature classification measures along automated systems allows these constraints to decrease.

Overcoming Challenges:
The remedy to these challenges depends on maintaining precise ongoing interaction with stakeholders. Involving end-users through collaborative tools during the validation process enables the system to measure up to stakeholder requirements. Qualified analysts who handle requirements. Qualified analysts who handle requirement documentation and management decrease ambiguity in the process.

Expanding SDLC Considerations

Agile Methodology in Healthcare Management

The combination of rapid development stages together with collaborative teamwork found in Agile methodology brings value to SDS model analyses for HMS systems. Healthcare administrators need to define telemedicine integration and COVID-19 vaccination management as top priorities for individual development cycles within their projects.

Advantages of Agile for HMS:

- Enhanced collaboration with stakeholders.
- Faster delivery of functional components.
- Using Agile provides the healthcare sector improved capabilities to fulfil emerging healthcare needs.

Disadvantages:

- Requires consistent involvement from stakeholders.
- Large-scale regulatory compliance presents challenge to their operations.

Integration of DevOps Practices

The implementation of DevOps practices within the SDLC improves the efficiency of HMS deployment together with maintenance operations. For HMS a smoother operation leads to shorter periods of system inactivity and satisfied users.

Benefits of DevOps in HMS:

- Continuous delivery and integration.
- Improved system reliability and uptime.
- Faster response to user feedback.

Addressing Ethical Considerations

The implementation of ethical considerations takes a central role for healthcare systems. The SDLC process needs to address three ethical matters concerning patient data protection together with patient consent administration and fair healthcare access throughout all development stages.

Strategies for Ethical Compliance:

- Performing ethical impact assessments.
- Engaging diverse stakeholder groups in decision-making.
- Implementing strongly made data anonymization techniques.

Integration with Emerging Technologies

Artificial Intelligence (AI)

AI tools studying patient information help healthcare providers identify health dangers so they can set preventive measures before illnesses progress. Better healthcare quality combined with improved resource management emerges from the implementation of these systems. The introduction of AI technologies to healthcare operations by medical providers can generate improved treatment accuracy for specific patients which produces better healthcare delivery methods.

Applications of AI in HMS:

- The system performs automated work processes which include both appointment setup billing management tasks.
- Enhancing diagnostics through image recognition.
- Supporting clinical decision-making with real-time data insights.

Internet of Things (IoT)

Through IoT integration with HMS hospital can perform real-time patient health monitoring while providing better care services as well as remote medical consultations. The utilization of IoT sensors enables real-time monitoring of medical supplies through their tracking system.

Applications of IoT in HMS:

- Healthcare facilities depend on wearable remote patient monitoring solutions.
- The system provides automatic notification systems that identify significant health problems.
- Efficient management of hospital resources.

Block chain for Data Security

Block chain technology establishes a protected system to handled patient data storage together with data protection features against unauthorized data access. By implementing Block chain in HMS, it creates a trust between stakeholders.

Applications of Block chain in HMS:

- The system allows healthcare providers to exchange data through connected networks security.
- The system can enable auditing processes to monitor access to confidential patient data.

Additional SDLC Models and Frameworks

Prototyping Model

The Prototyping Model enables teams to construct an initial software version for getting user responses to make better requirements decisions. The Prototyping Model functions best when user needs remain unknown during the initial stage.

Advantages:

- The development process finds issues and gaps at an stage to save time and money when fixing them.
- Product development suffers less expensive changes because prototypes help refine requirements at an early stage.

Disadvantages:

- Ineffective management can turn the prototype cycle into a process that extends project scope from its planned path.
- Developing many prototypes costs an organization extra money if teams do not set proper controls.

Application to HMS:
Prototyping can be used as way to validate user interfaces, workflows and some important feature that we can not only observe or test but validate the user interfaces and workflows and some important feature we want to support and embrace Patient record management. This drives the requirement that end system closely aligns with the user needs. User feedback incorporated into the design and system functionality allows developers to refine the design and functionality so to achieve high user satisfaction and system efficiency.

V-Model (Validation and Verification Model)

Since it stresses strict validation and verification in each stage of the development, the V model is well suited to projects giving very high priority to quality and compliance.

Advantages:

- Strong focus on quality assurance.
- The phases of development have clear links to the stages of testing.
- Ideal for regulatory compliance in healthcare.

Disadvantages:

- The plan remains difficult to change once the development process starts.
- Each development phase in SDLC requires a significant amount of time than in Agile or incremental approaches.

Application to HMS:
The V-Model will help to ensure that all requirements around safety of patients and security of data for the HMS are both rigorously tested and validated. With this structured approach, this model reduces the likelihood of missing critical requirements as well as ensuring that the final system is reliable and meeting healthcare regulations.

Conclusion

Through the comparative analysis the Spiral Model is suited for developing the Healthcare Management System as it gives much emphasis on the process of risk management and adaptability to the changes. The next closet would be the Incremental Development Model for projects with similarly less complexity. For small and well defined Health Care Systems, the Waterfall Model could be used.

The detailed requirements engineering process is carried out in order to fulfil the HMS functional and non-functional requirements as well as address the potential risks and challenges. With the inclusion of Agile, DevOps, and ethical considerations, the strategy for the development grows further, first developing an effective system, and secondly, a socially responsible one.

The emerging technologies can be integrated into the HMS to optimize the level of success of the HMS, which include IoT, AI and Block chain to improve the accuracy, security of the use of HMS in patients and healthcare providers. By joining powerful tech with a powerful dev process, the system is built in a way to keep itself updated to upcoming challenges in health industry and keep helping it developing.

Appendix

Requirements Management tools:

- **JIRA:** functions as a complete solution through which teams organize their sprints while they track software defects from the beginning to the end of development.
- **IBM Engineering Requirements Management DOORS:** it functions as a solution for end-to-end requirements tracing to meet regulatory needs.
- **Trello:** It provides flexible project management through visual organization which improves team working efficiency.

Glossary

1. **HIPAA:** Health Insurance Portability and Accountability Act, a U.S. regulation for patient data privacy and enables smooth health data sharing among different treatment providers.
2. **EHR:** Electronic Health Record, functions as an extensive secure medical records database which contains complete patient medical history.
3. **SDLC:** Software Development Lifecycle, the process of planning, it is something that helps teams go through the journey of building reliable software.

References

1. Somerville, I. (2015). Software Engineering. Pearson.
2. Boehm, B. (1988). A spiral model of software development and enhancement. ACM SIGSO.

Shetty

ORIGINALITY REPORT

3%	2%	1%	1%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	www.coursehero.com Internet Source	1%
2	Submitted to Southern New Hampshire University - Continuing Education Student Paper	<1%
3	hk.aconf.cn Internet Source	<1%
4	nmamit.nitte.edu.in Internet Source	<1%
5	epdf.tips Internet Source	<1%
6	Zhaoxu Yang, Yifan Shao, Ye Wei, Jun Li. "Precision-Boosted Forest Fire Target Detection via Enhanced YOLOv8 Model", Applied Sciences, 2024 Publication	<1%

Exclude quotes	Off	Exclude matches	Off
Exclude bibliography	Off		